

Nonlinear Control Design for an Unmanned Aerial Vehicle for Path Following

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Abstract: In this work, we design a nonlinear controller for an autonomous unmanned aerial vehicle (UAV) to follow a predefined trajectory. The UAV is made to follow the desired path by driving the UAV's relative distance to the path, and its look angle to zero. The proposed design is easy to implement as it does not need path curvature information and uses the philosophy of target pursuit to follow the predefined path. We further demonstrate the merits of the proposed method through simulations for various cases in accurately tracking straight line and curvilinear paths.

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1. INTRODUCTION

One of the most rapid technological advancements in the past century segregates itself around autonomous vehicles, especially unmanned aerial vehicle (UAV) developments. From its early applications limited to the military, the scope of UAVs has expanded its horizons to other fields like surveillance, agriculture, photography, logistics, open field exploration, search and rescue missions, aerial filming, and packages delivering (Candiago et al., 2015; Chmaj and Selvaraj, 2015; Ma'Sum et al., 2013; Shakhathreh et al., 2019). In almost every scenario that a UAV encounters, perhaps precise trajectory tracking is the most fundamental and essential requirement. In this respect, the requirement for precise and accurate trajectory tracking algorithms has increased rapidly (Perez-Leon et al., 2019; Wang and Chen, 2016; Zhao et al., 2020), and forms the basis of our motivation.

In spite of several techniques proposed in the literature for trajectory tracking by a UAV, the most common methods are proportional-integral-derivative (PID) and linear quadratic regulator (LQR). However, these classical techniques do not always guarantee satisfactory results since they are linear in nature whereas a UAV has a highly nonlinear and underactuated dynamics. Katigbak et al. (2015) presented a study on trajectory tracking for a UAV using the classical PID controller. Although the controller showed good control performance and was easy to implement, the performance was not robust to system nonlinearities. Farag (2020) proposed a technique to overcome the insufficiency of PID controller by tuning its hyperparameters. This study presented three different design approaches to fine-tune the controller hyperparameters of a PID controller that facilitated track maneuvering for autonomous driving. The study in (Farag, 2020) used

various tuning methods, such as WAF-Tune, Twiddle algorithm, and Ziegler-Nichols approaches. However, these approaches required different parameters for different speeds. Another widely used optimal control method in the context of trajectory tracking is LQR wherein a quadratic cost function is optimized to obtain optimal control gains. Note that to implement an LQR algorithm, a nonlinear system (for example, a UAV), needs to be linearized around the trim conditions (Hernandez-Martinez et al., 2015; Ratnoo et al., 2011).

To deal with limitations associated with linear controllers, a nonlinear control method based on geometric relations was presented by Ha et al. (2019). In the work of Ha et al. (2019), an autonomous trajectory tracking with collision avoidance was introduced for multi-copters. The proposed method was designed as a combination of tracking control and collision avoidance control. Some researchers have also used backstepping control technique and its variants for attitude control and tracking of various paths. Gavilan et al. (2015) presented trajectory tracking for fixed-wing UAVs using model predictive control (MPC) and adaptive backstepping. A two-level control architecture for the guidance and control system was designed wherein the longitudinal controller was based on an adaptive backstepping scheme whereas the lateral directional control utilized LQR with an integral term.

Huang et al. (2019) presented a robust adaptive sliding mode control scheme for attitude and altitude tracking of a UAV considering the simultaneous effect of parametric uncertainties and external disturbances. The sturdiness of this method was further improved by additional adaptive laws to deal with the parametric uncertainties. Ren and Beard (2004) proposed a controller for a fixed-wing UAV considering constraints on UAV's velocity and the heading rate. The controller was designed using control

Lyapunov function (CLF) approach. CLF was used for the input constrained case to obtain the feasible sets of inputs that guaranteed the convergence of error along with optimization of an auxiliary performance function. Guo et al. (2012) offered an efficient strategy for an automatic steering controller consisting of feedback and a feedforward parts implemented on a prototype unmanned vehicle, DUTIV-1.

In the light of recent results, and to ensure an accurate trajectory tracking behavior for an autonomous vehicle, we propose a control law based on the position and the heading errors of an autonomous vehicle. One of the advantages of the proposed design lies in the fact that the reference path curvature is not needed, thereby making the proposed design simple and easy to implement. Another advantage of the proposed controller stems from the design similarities with pursuit guidance that needs simpler mechanization as compared to other advanced guidance strategies (Kumar and Ghose, 2015; Kumar and Mukherjee, 2019; Sinha and Kumar, 2020; Sinha et al., 2021a,b,c). Moreover, controller ensures that the autonomous vehicle tracks the desired path irrespective of its initial geometry with respect to the path.

The remainder of this paper is organized as follows. After an overview in Section 1, UAV model and the problem statement are presented in Section 2. The proposed control technique is designed in Section 3, followed by simulations in Section 4. Finally, Section 5 presents concluding remarks and directions for future investigation.

2. UAV MODEL AND PROBLEM STATEMENT

Consider a UAV moving with a constant linear speed, v_m , as shown in Fig. 1. We assume that the UAV is a point mass vehicle and that the motion of the UAV is restricted to a horizontal plane, which means the altitude of the UAV is constant. If the current position of the UAV is represented by $P(x_m, y_m)$ in Cartesian coordinates, then we can write the kinematics of UAV as

$$\dot{x}_m = v_m \cos \psi_m, \quad (1a)$$

$$\dot{y}_m = v_m \sin \psi_m, \quad (1b)$$

$$\dot{\psi}_m = \omega_m, \quad (1c)$$

where ψ_m and ω_m represent the heading angle and the heading angular rate of the UAV, respectively.

Let $S(x_r, y_r)$ is a reference point on the predefined path moving with a velocity, v_r , and heading angle, ψ_r . Similar to the kinematics of the UAV, the equations of motion of $S(x_r, y_r)$ can be written as

$$\dot{x}_r = v_r \cos \psi_r, \quad (2a)$$

$$\dot{y}_r = v_r \sin \psi_r, \quad (2b)$$

$$\dot{\psi}_r = \omega_r, \quad (2c)$$

where $\dot{x}_r$ and $\dot{y}_r$ represent the velocity of the point S in x and y directions, while ω_r represent the heading angular rate of the reference point, S .

In regards to Fig. 1, the relative distance between the UAV and the reference point is denoted by r , while the line-of-sight (LOS) angle of the UAV with respect to the reference point is θ . For convenience, we define an angle, ξ , as the difference between ψ_m and θ . In order to track

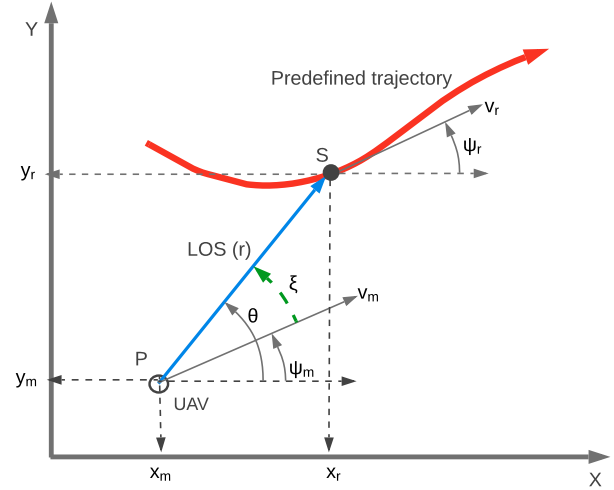


Fig. 1. Schematic representation of path tracking problem.

the reference path, the UAV is steered using its linear and angular velocities.

In this paper, our aim is to make the UAV track an arbitrary reference path in the absence of path curvature information. Another goal is to make the design independent of the UAV's relative geometry with respect to the path. Notice that the kinematics equations for the UAV and the reference path are nonlinear, and it is our intention to treat the problem intact in its nonlinear form. This will allow the proposed controller to remain effective for a wider domain.

3. MAIN RESULTS

The UAV's mission is to track the desired path. A part of this requirement entails nullifying the UAV's relative position error with respect to the reference point on the path. Consider the relative distance between the UAV and the reference point, r , which can be written as

$$r = \sqrt{(x_r - x_m)^2 + (y_r - y_m)^2}, \quad (3)$$

whose time derivative can be computed as

$$\dot{r} = \frac{(x_r - x_m)(\dot{x}_r - \dot{x}_m) + (y_r - y_m)(\dot{y}_r - \dot{y}_m)}{r}. \quad (4)$$

On substituting (1) and (2) in (4), one may obtain

$$\dot{r} = (v_r \cos \psi_r \cos \theta - v_m \cos \psi_m \cos \theta) + (v_r \sin \psi_r \sin \theta - v_m \sin \psi_m \sin \theta) \quad (5)$$

which can be further simplified to

$$\dot{r} = v_r \cos(\theta - \psi_r) - v_m \cos(\theta - \psi_m). \quad (6)$$

Using geometrical relations, the LOS angle, θ , can be written as

$$\theta = \tan^{-1} \frac{y_r - y_m}{x_r - x_m} \quad (7)$$

whose time derivative may be expressed as

$$\dot{\theta} = \frac{-v_r \sin(\theta - \psi_r) + v_m \sin(\theta - \psi_m)}{r}. \quad (8)$$

Note that if the UAV points in the direction of the LOS, and reduce its relative distance with respect to the reference point on the path, then it is certain that it will

eventually reach the reference point provided $v_m > v_r$. Therefore, it is apparent to drive the angle,

$$\xi = \theta - \psi_m \quad (9)$$

to zero, thereby bringing the UAV on a pursuit course to the reference point. To steer on a pursuit course to the reference point, it is essential for the UAV to match its heading angular rate with the rate of turn of the LOS. In essence, the UAV now *pursues*, or carries out a *tail-chase* motion to the reference point, S .

In order for the UAV to reach the reference point on the path, we design a controller to nullify r . This is presented in the next theorem.

Theorem 1. Consider the equations of motion for the UAV and the reference point, given in (1)–(2), and the UAV's relative distance with respect to the reference point, r , in (3). If the UAV applies a linear velocity,

$$v_m = \frac{v_r \cos(\theta - \psi_r) + \mathcal{M}_0 r}{\cos(\theta - \psi_m)} \quad (10)$$

where $\mathcal{M}_0 > 0$ is the gain of the controller, then it approaches the reference point, S , asymptotically.

Proof. Let us define a Lyapunov function candidate, $V_1 = \frac{1}{2}r^2$. On differentiating V_1 with respect to time, and using (4), one may obtain

$$\dot{V}_1 = r\dot{r} = r(v_r \cos(\theta - \psi_r) - v_m \cos(\theta - \psi_m)). \quad (11)$$

With the linear velocity of the UAV given in (10), the expression in (11) becomes

$$\dot{V}_1 = -\mathcal{M}_0 r^2 < 0 \quad \forall \mathcal{M}_0 > 0, r \neq 0, \quad (12)$$

implying that $r \rightarrow 0$ asymptotically whose rate of convergence is determined by $\mathcal{M}_0$. This concludes the proof. $\square$

Our next goal is to drive the angle, ξ , to zero in order for the UAV to steer the UAV on a pursuit course to S . This is presented next.

Theorem 2. Consider the equations of motion for the UAV and the reference point, given in (1)–(2), and the UAV's look angle, ξ , in (9). If the UAV applies a heading angular velocity,

$$\omega_m = \frac{v_r}{r} (\tan(\theta - \psi_m) - \sin(\theta - \psi_r)) + \mathcal{M}_0 r \sin(\theta - \psi_m) + \mathcal{N}_0 \xi \quad (13)$$

where $\mathcal{N}_0 > 0$ is the gain of then controller, then it approaches the pursuit course to the reference point asymptotically.

Proof. Let us define a Lyapunov function candidate, $V_2 = \frac{1}{2}\xi^2$, whose time differentiation results in

$$\dot{V}_2 = \xi \dot{\xi} = \xi (\dot{\theta} - \dot{\psi}_m). \quad (14)$$

On substituting the expression of (8) in (14), with the fact that $\dot{\psi}_m = \omega_m$, one has

$$\dot{V}_2 = \xi \left(\frac{-v_r \sin(\theta - \psi_r) + v_m \sin(\theta - \psi_m)}{r} - \omega_m \right). \quad (15)$$

On substituting the heading angular velocity of the UAV given in (13), the above expression reduces to

$$\dot{V}_2 = -\mathcal{N}_0 \xi^2, \quad (16)$$

which is negative for all $\mathcal{N}_0 > 0$ and $\xi \neq 0$. Therefore, it can be inferred that V_2 , and hence ξ , approaches zero asymptotically whose rate of convergence depends on $\mathcal{N}_0$. This completes the proof. $\square$

Remark 1. If ξ is zero initially, then it means that the UAV is already on the pursuit course towards the reference point.

Remark 2. Observe from Theorem 2 that r appears in the denominator of (13). However, this does not cause any issues during implementation since the term $(\theta - \psi_r)$ vanishes earlier than r could become zero. As a matter of fact, ξ also vanishes in accordance with $(\theta - \psi_r)$. This further leads to an inference that the error between ψ_r and ψ_m gets nullified, which means that the UAV's heading is aligned with that of the reference point during path following.

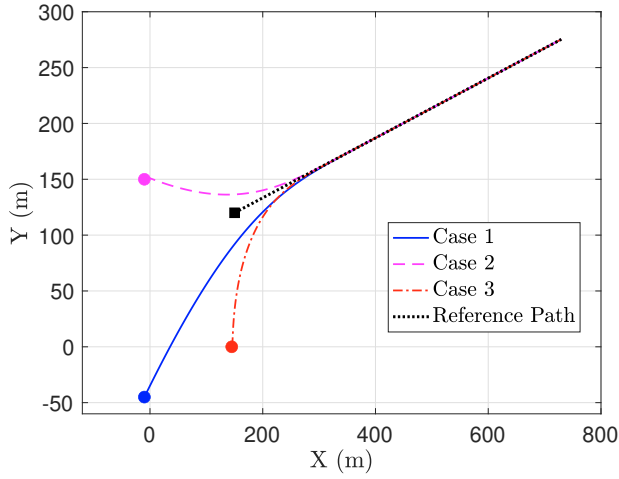
4. SIMULATIONS

In this section, we show the efficacy of the proposed path tracking control for a UAV under various path curvatures. We first consider a straight line path since it is one of the most important considerations in path following scenarios, and is an essential requirement in waypoint following. We, then, present simulations for a path with arbitrary curvature to show that the proposed design delivers satisfactory performance under various scenarios.

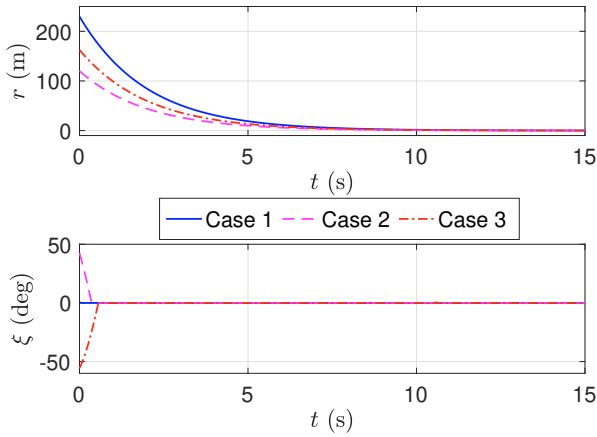
We assume that the reference point moves with a speed of 40 m/s with an initial heading angle of 15° . The initial heading angle of the UAV is 45° . The controller gains are chosen as $\mathcal{M}_0 = 0.5$, and $\mathcal{N}_0 = 45$. In the results that follow, the desired curve is described by $\sin(\pi t)$.

Figs. 2–3 depict three cases when the UAV starts from various initial locations (depicted using circle markers) and tracks the reference path within same time in each case. In different cases, the UAV starts from $P(-10, -45)$, $P(145, 0)$, and $P(-10, 150)$. This shows that even if the UAV starts far away from the reference path, it will eventually track the path. However, in such case, the UAV may take more time to converge to the desired path. One may also notice that the UAV's angular velocity reduces in magnitude earlier than the UAV's linear velocity in each case. This can be attributed to the fact that the UAV nullifies its look angle error first to pursue the time-varying reference point on the desired path. Overall, it is also observed that the control demand reduces when the UAV has converged to the desired path.

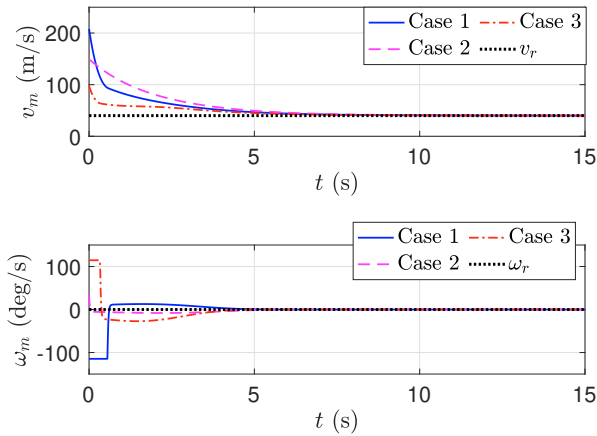
We compare the results of this paper with those presented in the work of Ha et al. (2019), as shown in Figs. 4–5. It is observed that the proposed controller allows the UAV to converge to the desired paths earlier than the controller presented in (Ha et al., 2019) with reasonable control requirements. Figs. 4–5 depict scenarios when the UAV is made to follow straight line and curvilinear paths. We consider a straight line path joining $P(70, 90)$ and $S(647.12, 244.6)$ whose slope is 0.267 in the Cartesian plane. The UAV starts from $(50, -10)$, depicted in the trajectory plots (see Figs. 4a–5a) using a blue circle marker. The initial position of the reference point is represented using a red square marker. It is evident from



(a) UAV's trajectories and the reference path.

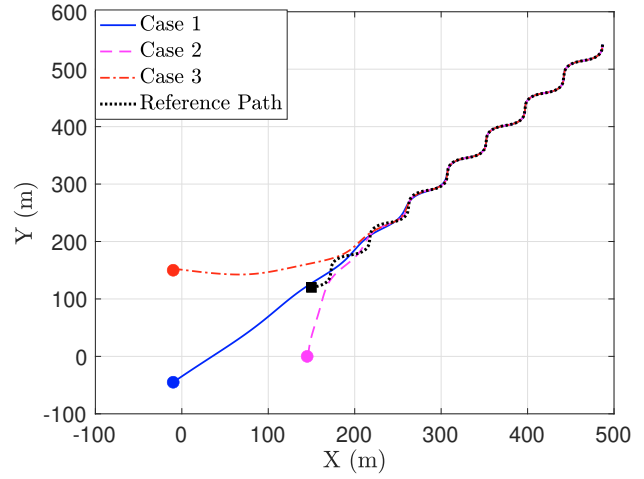


(b) UAV's relative distance to the path, and its look angle.

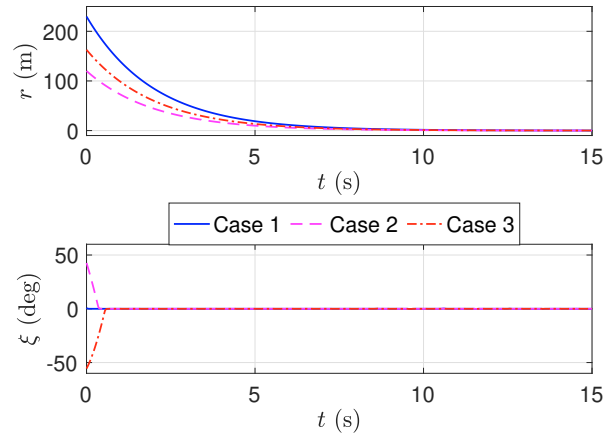


(c) UAV's linear and angular velocities.

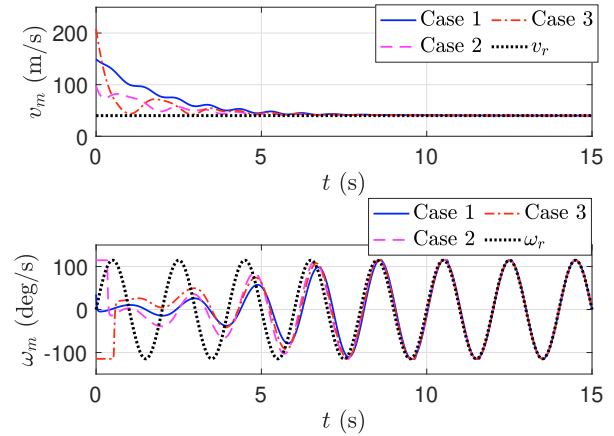
Fig. 2. Straight line path following for various UAV start locations.



(a) UAV's trajectories and the reference path.



(b) UAV's relative distance to the path, and its look angle.

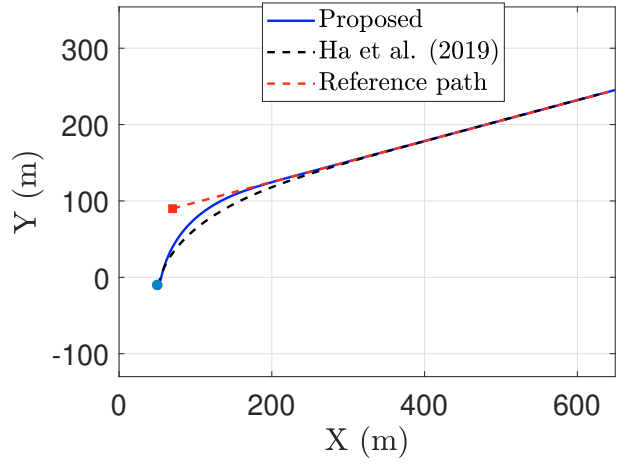


(c) UAV's linear and angular velocities.

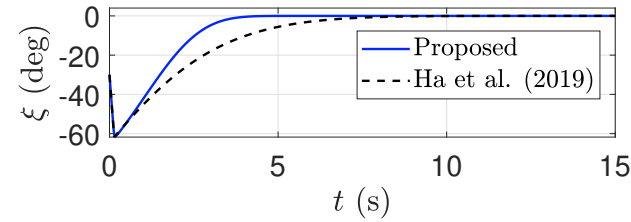
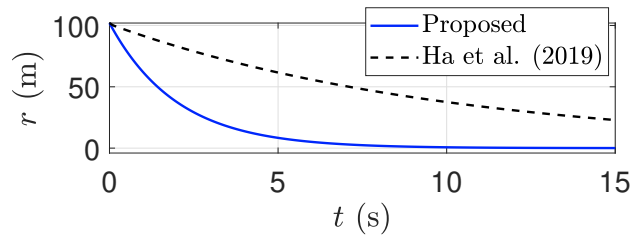
Fig. 3. Curvilinear path following for various UAV start locations.

the trajectory plots in Figs. 4–5 that the UAV is able to follow the desired paths.

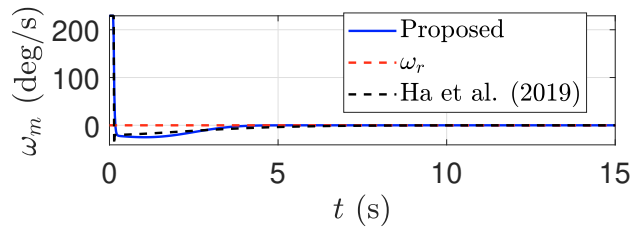
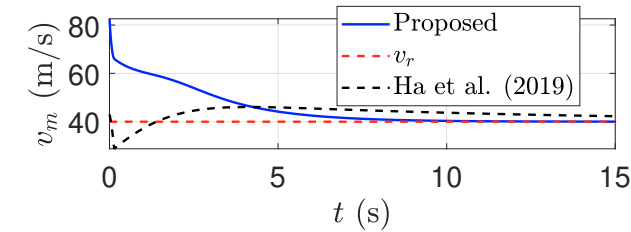
The control demand in each case matches the linear and angular velocities of the reference point in approximately same time irrespective of the curvature of the trajectory.



(a) UAV's trajectories and the reference path.

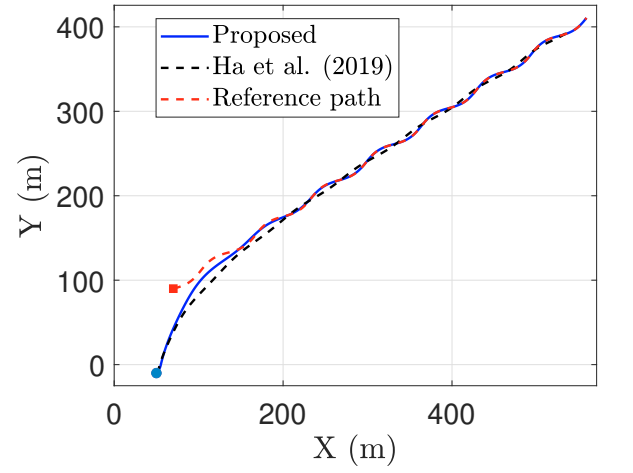


(b) UAV's relative distance to the path, and its look angle.

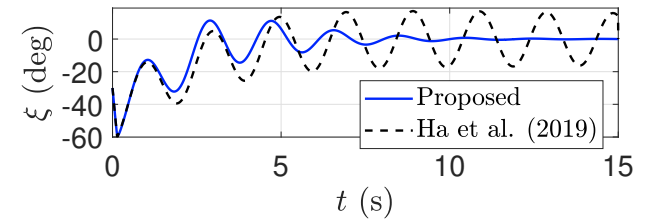
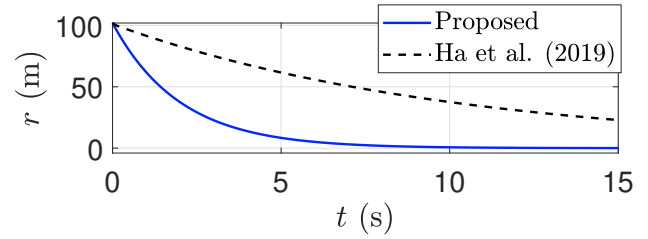


(c) UAV's linear and angular velocities.

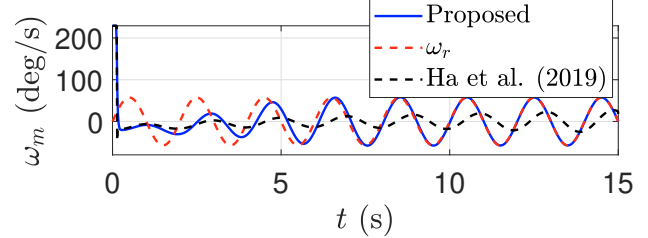
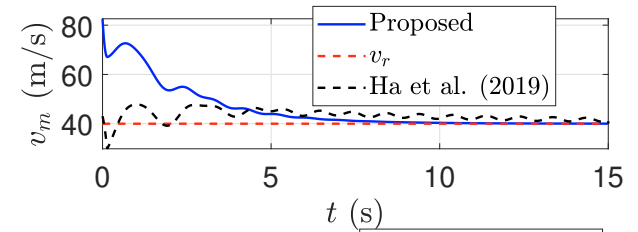
Fig. 4. Comparison with an existing strategy for straight line path following.



(a) UAV's trajectories and the reference path.



(b) UAV's relative distance to the path, and its look angle.



(c) UAV's linear and angular velocities.

Fig. 5. Comparison with an existing strategy for curvilinear path following.

5. CONCLUSIONS

It is also observed that the requirement of control demand at the initial state is also similar in both cases.

In this paper, we designed a simple and easy-to-implement control strategy for a UAV to follow a desired path using

the philosophy of pursuit guidance. By driving the UAV's relative distance and look angle to zero, the UAV was made to converge to the desired path without needing the path curvature information. We also presented a comparison between the proposed technique and an existing one to further demonstrate the efficacy of the present work. Making the UAV traverse through obstacles may be an interesting future work.

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REFERENCES

- Candiago, S., Remondino, F., De Giglio, M., Dubbini, M., and Gattelli, M. (2015). Evaluating multispectral images and vegetation indices for precision farming applications from uav images. *Remote sensing*, 7(4), 4026–4047.
- Chmaj, G. and Selvaraj, H. (2015). Distributed processing applications for uav/drones: a survey. In *Progress in Systems Engineering*, 449–454. Springer.
- Farag, W. (2020). Complex trajectory tracking using pid control for autonomous driving. *International Journal of Intelligent Transportation Systems Research*, 18(2), 356–366.
- Gavilan, F., Vazquez, R., and Esteban, S. (2015). Trajectory tracking for fixed-wing uav using model predictive control and adaptive backstepping. *IFAC-PapersOnLine*, 48(9), 132–137.
- Guo, J., Hu, P., Li, L., and Wang, R. (2012). Design of automatic steering controller for trajectory tracking of unmanned vehicles using genetic algorithms. *IEEE Transactions on Vehicular Technology*, 61(7), 2913–2924.
- Ha, L.N.N.T., Bui, D.H.P., Hong, S.K., et al. (2019). Nonlinear control for autonomous trajectory tracking while considering collision avoidance of uavs based on geometric relations. *Energies*, 12(8), 1551.
- Hernandez-Martinez, E.G., Fernandez-Anaya, G., Ferreira, E., Flores-Godoy, J.J., and Lopez-Gonzalez, A. (2015). Trajectory tracking of a quadcopter uav with optimal translational control. *IFAC-PapersOnLine*, 48(19), 226–231.
- Huang, T., Huang, D., Wang, Z., and Shah, A. (2019). Robust tracking control of a quadrotor uav based on adaptive sliding mode controller. *Complexity*, 2019.
- Katigbak, C.N.R., Garcia, J.R.B., Gutang, J.E.D., De Villa, J.B., Alcid, A.D.C., Vicerra, R.R.P., Cruz, A.R.D., Roxas, E.A., and Serrano, K.K.D. (2015). Autonomous trajectory tracking of a quadrotor uav using pid controller. In *2015 International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment and Management (HNICEM)*, 1–5. IEEE.
- Kumar, S.R. and Ghose, D. (2015). Impact time guidance for large heading errors using sliding mode control. *IEEE Transactions on Aerospace and Electronic Systems*, 51(4), 3123–3138.
- Kumar, S.R. and Mukherjee, D. (2019). Deviated pursuit based interception at a priori fixed time. *Journal of Guidance, Control, and Dynamics*, 42(9), 2124–2131.
- Ma'Sum, M.A., Arrofi, M.K., Jati, G., Arifin, F., Kurniawan, M.N., Mursanto, P., and Jatmiko, W. (2013). Simulation of intelligent unmanned aerial vehicle (uav) for military surveillance. In *2013 international conference on advanced computer science and information systems (ICACSIS)*, 161–166. IEEE.
- Perez-Leon, H., Acevedo, J.J., Millan-Romera, J.A., Castillejo-Calle, A., Maza, I., and Ollero, A. (2019). An aerial robot path follower based on the 'carrot chasing' algorithm. In *Iberian Robotics conference*, 37–47. Springer.
- Ratnoo, A., Sujit, P., and Kothari, M. (2011). Adaptive optimal path following for high wind flights. *IFAC Proceedings Volumes*, 44(1), 12985–12990.
- Ren, W. and Beard, R.W. (2004). Trajectory tracking for unmanned air vehicles with velocity and heading rate constraints. *IEEE Transactions on Control Systems Technology*, 12(5), 706–716.
- Shakhatreh, H., Sawalmeh, A.H., Al-Fuqaha, A., Dou, Z., Almaita, E., Khalil, I., Othman, N.S., Khreishah, A., and Guizani, M. (2019). Unmanned aerial vehicles (uavs): A survey on civil applications and key research challenges. *IEEE Access*, 7, 48572–48634.
- Sinha, A. and Kumar, S.R. (2020). Supertwisting control-based cooperative salvo guidance using leader-follower approach. *IEEE Transactions on Aerospace and Electronic Systems*, 56(5), 3556–3565.
- Sinha, A., Kumar, S.R., and Mukherjee, D. (2021a). Non-singular impact time guidance and control using deviated pursuit. *Aerospace Science and Technology*, 115, 106776.
- Sinha, A., Kumar, S.R., and Mukherjee, D. (2021b). Three-dimensional guidance with terminal time constraints for wide launch envelopes. *Journal of Guidance, Control, and Dynamics*, 44(2), 343–359.
- Sinha, A., Kumar, S.R., and Mukherjee, D. (2021c). Three-dimensional nonlinear cooperative salvo using event-triggered strategy. *Journal of Guidance, Control, and Dynamics*, 44(2), 328–342.
- Wang, H. and Chen, M. (2016). Trajectory tracking control for an indoor quadrotor uav based on the disturbance observer. *Transactions of the Institute of Measurement and Control*, 38(6), 675–692.
- Zhao, Z., Cao, D., Yang, J., and Wang, H. (2020). High-order sliding mode observer-based trajectory tracking control for a quadrotor uav with uncertain dynamics. *Nonlinear Dynamics*, 102(4), 2583–2596.